

Engine and Validation

Continuum Engine and Validation Protocol

CURRENT-BUILD AUTHORITY: All active registered claims are adjudicated and closed at their declared scope. Terminal classes described in this document are execution semantics for posed inputs and do not denote unfinished canonical project work.

Canonical closure interpretation — upstream resolution required

A registered claim is certified closed when the record states **what SEAM resolves upstream that produces the observation or makes the conventional description downstream**. The required order is:

native complete structure → native relation/field/selector consequence → frozen result → conventional representation/comparator

For each registered claim the authoritative row-level linkage is `SEAM_210_CLAIM_CERTIFICATION_REGISTER.md` and the claim-specific record under `Evidence/Claim_Certification/`. Those records name the upstream mechanism, sealed execution/result, quantitative standing where applicable, scientific comparator, and exact boundary.

No-reduction rule (see `00_README.md`). Structure in this archive is retained faceted and is never collapsed to a scalar. A faceted or symbolic form is a complete description of a resolved structure, not a deferred value. It does not license reading an unevaluated row as resolved: see the three-state status vocabulary (`RESOLVED` / `EXCLUDED` / `NOT_EVALUABLE`) in `00_README.md`.

Archive: Continuum Paradigm

Lineage: ESIH → ESAM → SEAM → Continuum → Manifold

Role: Executable architecture, Q-ARCV/M-ARCV/A-ARCV discipline, cold-reader engineering protocol, runner boundaries, transcript requirements, and scoring rules.

Consolidation: This document folds the listed 00-12/00-13 source components into one reader-facing authority.

Canonical architecture and execution scope

The Continuum Paradigm uses one integrated architecture with distinct responsibilities:

ESIH = definition of permissible structure and possible existence

AEPS = definition of how energy interacts with and distributes across structure

SEAM = mathematical representation and execution of structure, relation, interaction, and state change

ESAM is the mathematical-formalization lineage through which the ESIH structural principles are expressed in the SEAM mathematics.

The narrative definition states the physical meaning of a principle. The formalism supplies the complete mathematical state required to calculate that principle in a particular case:

formalism = mathematical realization of the narrative definition

The existence standard distinguishes possible and realized existence:

$\mathcal{C}_{\text{possible}} = \mathcal{A}_{\text{SEAM}}$

$\mathcal{C}_{\text{realized}} \subseteq \mathcal{A}_{\text{SEAM}}$

The Continuum is the physical domain. The Continuum Manifold is the retained and normalized record of observed or resolved portions of that domain:

$\mathcal{M}_t = \text{RetainNormalize}(\text{ObservedResolved}(\mathcal{C}_{\text{realized}}))$

Current quantitative execution scope

Architectural scope and executed numerical scope are recorded separately.

The atomic structural domain carries the fully explicit numerical admissibility implementation presently retained in the archive. Molecular, aggregate, electromagnetic, propagation, material, biological, and macroscopic domains inherit the same structural admissibility and interaction architecture through their declared native constructors.

Chemistry and biology presently carry architecture-level structural coverage. Domain-wide quantitative completion enters the evidence standing through executed native calculations.

Bond interaction architecture is retained where documented. Numerical bond lengths and dissociation energies enter the evidence standing when their native calculation, seal, and comparator record are present.

Macroscopic interaction and force architecture are retained where documented. Absolute SI force values and a numerical derivation of G enter the evidence standing through their own completed native execution and evidence records.

Continued calculation expands quantitative coverage while preserving the same first-principles architecture.

Source components folded here

- `02_SEAM_CANONICAL_ENGINEERING_SPEC.md`
- `COLD_READER_ENGINEERING_PROTOCOL_v1_1.md`

Part 1: Consolidated Source —

`02_SEAM_CANONICAL_ENGINEERING_SPEC.md`

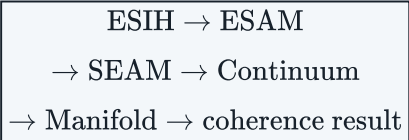
SEAM Canonical Engineering Specification

System: Continuum Paradigm; SEAM — Systemic Empirical Atomic Model is the causal substrate layer
Standing: Complete
Document role: Concise engineering definition of the completed canonical system.
Mathematical authority: `02_CONTINUUM_TECHNICAL_FOUNDATIONS.md`

1. System definition

SEAM is the executable mathematical layer of a structure-first physical framework.

The canonical dependency stack is:



The system resolves causal physical structure first and expresses conventional observables afterward.

Layer responsibilities

Layer	Engineering responsibility	Output
ESIH	First-principle structural constraints	Admissible primitive/constraint set
ESAM	Mathematical realization	Structural state, configuration, entropy, coupling, closure mathematics
SEAM	Deterministic execution	Primitive result, closure trace, projection, validation state
Continuum	Persistent validated reference	Retained structural/observational state

Layer	Engineering responsibility	Output
Manifold	Active normalized comparison	Query-specific comparison structure
Resolver	Representation of resolved state	Human/machine-readable answer

2. Canonical physical ordering

The system executes the following causal order:

```

input → observation/primitive
      → structure → configuration
      → entropy selection → projection
→ structural coherence → Manifold comparison
      → validation → primitive result
                        → answer

```

The complete closure operator is:

$$\text{Closure}_{SEAM}(X) = VC(MC(CC(TC(SC(\mathfrak{R}(X)))))$$

where:

- $\mathfrak{R}$ = representation constructor;
- SC = structural/configuration closure;
- TC = transfer/trajectory stage;
- CC = internal structural/continuity coherence;
- MC = Manifold comparison;
- VC = validation closure.

3. Primitive structural representation

3.1 Atomic baseline

The canonical atomic start is the electron-count baseline:

$$Z(n).$$

The seven-shell capacity vector is:

$$c_{shell} = [2, 8, 18, 32, 18, 32, 18].$$

3.1.1 Elemental shell-admissibility count

The seven-shell capacity vector gives the native positive-Z elemental shell-admissibility ceiling directly:

$$2 + 8 + 18 + 32 + 18 + 32 + 18 = 128.$$

Thus the shell-admissible positive-Z elemental baseline domain is:

$$Z = 1, 2, \dots, 128.$$

The shell-closure sequence is:

$$2, 10, 28, 60, 78, 110, 128.$$

The archive's 118-element results are the current empirical comparison and qualification range. They do not define the mathematical ceiling of the seven-shell architecture. The null $Z = 0$ state may be represented as an empty shell state for integer accounting, but it is not an elemental baseline.

For entity i and shell n :

$$0 \leq N_{i,n} \leq c_n, \quad \rho_{i,n} = \frac{N_{i,n}}{c_n}.$$

The entity shell state is:

$$\mathcal{S}_i = \{\rho_{i,1}, \dots, \rho_{i,7}\}.$$

3.2 Relational state

The complete relational state is:

$$\mathcal{R} = \{r_{ij}, \chi_{ijkl}\}.$$

with pair separation r_{ij} and reflection/chirality parity

$$\chi_{ijkl} = \text{sgn}[(r_j - r_i) \cdot ((r_k - r_i) \times (r_l - r_i))].$$

3.3 Complete configuration

$$C = (\{\mathcal{S}_i\}, \mathcal{R}).$$

The admissible configuration set $\mathcal{A}$ contains configurations satisfying the declared count, capacity, topology, relation, boundary, and contract constraints.

4. Entropy-directed configuration resolution

The governing selection rule is:

$$C^* = \arg \max_{C \in \mathcal{A}} S[C].$$

The canonical functional is:

$$S[C] = S_{config}[C] + \lambda_{field} S_{field}[C] + \lambda_{coupling} S_{coupling}[C].$$

4.1 Configuration entropy

$$S_{config}[C] = k_B \sum_i \sum_{n=1}^7 \ln \left(\frac{c_n}{N_{i,n}} \right).$$

4.2 Native shell support

For dimensionless native coordinate ξ and entity center X_i :

$$\Omega_{i,n}(C) = \{\xi : n-1 \leq \|\xi - X_i\| < n\}.$$

Shell volume:

$$V_n = \frac{4\pi}{3} [n^3 - (n-1)^3].$$

Normalized basis:

$$u_{i,n}(\xi | C) = \frac{\mathbf{1}_{\Omega_{i,n}(C)}(\xi)}{\sqrt{V_n}}.$$

4.3 Structural field

$$F_C(\xi) = \sum_i \sum_{n=1}^7 \sqrt{N_{i,n}} u_{i,n}(\xi | C).$$

Normalized density:

$$q_C(\xi) = \frac{|F_C(\xi)|^2}{\int |F_C(\xi)|^2 d^3\xi}.$$

Field entropy:

$$S_{field}[C] = -k_B \int q_C(\xi) \ln q_C(\xi) d^3\xi.$$

4.4 Shell-overlap operator

$$O_{ij}^{nm}(C) = \int \sqrt{|u_{i,n}|^2 |u_{j,m}|^2} d^3\xi = \frac{\text{Vol}(\Omega_{i,n} \cap \Omega_{j,m})}{\sqrt{V_n V_m}}.$$

4.5 Coupling entropy

$$S_{coupling}[C] = k_B \sum_{i < j} \sum_{n,m} \rho_{i,n} \rho_{j,m} O_{ij}^{nm}(C).$$

4.6 Single-atom shell-field evaluation identity

For a single atom, the complete field construction remains operative: $F_C(\xi)$ and the normalized density $q_C = |F_C|^2 / \int |F_C|^2 d^3\xi$ are not discarded. Canonical annular supports are disjoint for intra-atomic spatial cross-shell overlap, so

$$S_{coupling}^{spatial} \equiv 0$$

for that spatial cross-overlap term, while shared field normalization supplies the inter-shell field contribution. The expression

$$S_{field}(E) = \ln E - \frac{1}{E} \sum_n N_n \ln \frac{N_n}{V_n}$$

is an evaluation identity for the declared single-atom shell-field construction, not a replacement of the complete field definition. Its numerical agreement with the direct field integral is reported over the current empirical comparison record.

4.7 Signed shell-field operator

Define the causal/backward second-difference shell-response operator where two prior count states exist:

$$\mathcal{O}_n(k) = \Delta_-^2 S_{field}(E) = S_{field}(E) - 2S_{field}(E-1) + S_{field}(E-2), \quad E = B_n + k, \quad B_n = \sum_{m < n} c_m.$$

This operator is the canonical signed shell-field response on the complete architectural domain where two prior states exist: $Z = 3 \dots 128$, **126 evaluable states**. The current empirical-comparison subset $Z = 3 \dots 118$ contains **116 evaluable states**. Execution establishes positive response at evaluable shell openings and negative response under continued fill:

$$\mathcal{O}_n(1) > 0 \text{ where evaluable,} \quad \mathcal{O}_n(k) < 0 \quad (k > 1).$$

The qualification result is **0 violations over 126/126 architectural evaluable states**, including **0 violations over the 116/116 empirical-comparison subset**. The closed-form operator reproduces the direct field backward second difference to machine zero across the evaluable within-shell states.

The operator depends on total electron count E , shell volume V_n , and the causal/backward count stencil; these structural coordinates carry the canonical field response.

4.8 Canonical operator integration path

The shell-field operator is consumed as structural state within the full entropy and closure architecture:

$$\begin{aligned} Z(n) &\rightarrow \{N_n\} \\ &\rightarrow S_{field} \rightarrow \mathcal{O}_n(k) \\ &\rightarrow S[C] \rightarrow C^* \\ &\rightarrow \text{Closure}_{SEAM}(X) \rightarrow P_X. \end{aligned}$$

The invariant engineering object at the shell-response layer is $\mathcal{O}_n(k)$, while configuration selection remains governed by the complete three-term functional $S[C]$. The shell-response operator therefore contributes to, but does not replace, configuration entropy, inter-entity coupling, structural closure, Manifold comparison, or validation closure.

The composition boundary is:

$$\mathcal{O}_n(k) \subset S_{field}[C] \subset S[C] \xrightarrow{\arg \max} C^* \xrightarrow{\mathfrak{R}, SC, TC, CC, MC, VC} P_X.$$

For time-bearing projections, TC consumes the canonical Cs-resolved interval

$$T(X) = \frac{n_{Cs}(X)}{9,192,631,770}.$$

For neutron-conditioned atomic evidence, the resolved native state retains

$$\mathcal{N}_Z = [N_{low}, N_{high}], \quad M = M(Q_{res}, N),$$

so empirical adjudication preserves the complete native coordinate set through the final evidence join.

4.9 Entropy-resolution state evolution

The canonical configuration selector and the continuously evolving entropy-resolution state are related but distinct objects.

$$C^* = \arg \max_{C \in \mathcal{A}} S[C]$$

selects the admissible configuration under the canonical entropy law. The realized state then participates in the continuous entropy-resolution loop

$$\begin{aligned} C_n &\rightarrow \Xi_n \\ &\rightarrow \Delta C_n \rightarrow C_{n+1} \\ &\rightarrow \Xi_{n+1} \rightarrow \text{entropy-directed evolution.} \end{aligned}$$

Ξ is the current entropy-resolution state generated from the accepted structural/entropy content. It is not an independent admission gate and does not replace $S[C]$.

This distinction is load-bearing:

$$\text{native physical evolution} \rightarrow \text{projection} \rightarrow \text{Closure}_{SEAM}(X).$$

The operational closure composition validates, compares, and resolves the represented state downstream of native evolution.

5. Single interaction law

The governing structural interaction is common across construction scale:

$$\boxed{F_{int}^{atomic} \equiv F_{int}^{molecular} \equiv F_{int}^{composite} .}$$

The identity denotes a common governing law. Each realized configuration supplies its own state, geometry, magnitude, and projection.

5A. Variational-field interaction and macroscopic projection

For a pair configuration evaluated at native separation N_r , the canonical interaction path resolves the admissible field state from the same entropy functional that selects the physical configuration.

5A.1 Field-state resolution

For each occupied shell,

$$\Omega_{i,n}(C, N_r) = \{\xi : n - 1 \leq \|\xi - X_i\| < n\},$$

and the admissible internal-field class is

$$U_{i,n}(C, N_r) = \{u \geq 0 : \text{supp } u \subseteq \Omega_{i,n}, \int_{\Omega_{i,n}} |u|^2 d^3\xi = 1\}.$$

The uniform indicator

$$u_{i,n}^{(0)} = \mathbf{1}_{\Omega_{i,n}} / \sqrt{V_n}$$

is the isolated-state/fixed-branch baseline; it is not imposed as the universal interacting profile.

At fixed N_r , the selected field is

$$\{u_{i,n}^*(N_r)\} = \arg \max_{\{u_{i,n}\} \in \prod U_{i,n}(C, N_r)} S[C; N_r, \{u\}].$$

Shell support, occupancy, topology, entropy composition, coupling law, and the comparator firewall remain frozen. The field may redistribute only within its already-declared shell support. No target-derived profile or post-run tuning is admissible.

The resolved field objects are

$$F_C(\xi; N_r) = \sum_{i,n} \sqrt{N_{i,n}} u_{i,n}^*(\xi; N_r),$$

$$q_C = |F_C|^2 / \int |F_C|^2 d^3\xi, \quad p_{i,n} = |u_{i,n}^*|^2,$$

$$O_{ij}^{nm}(N_r) = \int \sqrt{p_{i,n} p_{j,m}} d^3\xi.$$

The separation-dependent selected entropy is

$$S^*(N_r) = S[C^*; N_r, \{u^*(N_r)\}].$$

5A.2 Pair-energy consequence and force

The native radial response is

$$R_{\text{SEAM}}(N_r) = S^*(N_r).$$

When a run contract selects a finite separation,

$$N_r^* = \arg \max_{N_r} S^*(N_r),$$

with the declared interior-extremum conditions applied to the complete selected state.

The resolved state is carried into the selected-state energy representation:

$$\Delta H_{XY}^{(J)}(N_r) = E_H[C_{XY}^*(N_r)] - E_H[C_X^* \oplus C_Y^*].$$

The Hamiltonian/Joule bridge is

$$H^{(J)} = E_0 \tau_{\text{SEAM}} + \int T dS.$$

The pair force follows from the resolved energy curve:

$$F_{XY}(N_r) = -\frac{1}{L_{XY}} \frac{d\Delta H_{XY}^{(J)}(N_r)}{dN_r}, \quad R = L_{XY} N_r.$$

The canonical dependency is therefore

$$\begin{aligned} C^* &\rightarrow \{u^*(N_r)\} \\ &\rightarrow S^*(N_r) \rightarrow E_H[C^*(N_r)] \\ &\rightarrow \Delta H_{XY}^{(J)}(N_r) \rightarrow F_{XY}(N_r). \end{aligned}$$

A native electric/magnetic/mixed-channel decomposition and an `eta(N_r)/N_r` attraction kernel are not inputs to the **v18.6 variational molecular selector**. The long-range attraction/long-range attraction branch has its own separately declared H_{attr} constructor using `eta(N_r)/N_r`; executors must not erase that branch or import the v18.6 `TAU_PAIR_MAPPING_UNBOUND` terminal into it.

5A.3 Scale continuation

The pair result retains the same governing entropy law used for atomic and molecular resolution. Finite-body and macroscopic projections aggregate resolved pair and field structure while preserving the native dependency chain. The resulting macroscopic attraction is therefore an observable projection of the entropy-selected SEAM field state.

This section is part of the canonical interaction law and preserves one foundational interaction class across scale.

6. Structural trajectory

A state trajectory is represented as:

$$C_0 \rightarrow C_1 \\ \rightarrow \dots \rightarrow C_k.$$

Each transition records:

- predecessor state;
- applied entity/change/distribution/modification;
- resulting state;
- declared operator;
- preserved invariants;
- structural differential;
- closure state;
- time binding when physical evolution is represented.

The entropy-directed evolution loop is:

$$C_n \rightarrow \Xi_n \\ \rightarrow \text{applied change} \rightarrow C_{n+1} \\ \rightarrow \Xi_{n+1} \rightarrow \text{entropy-directed response}.$$

7. Metrology

7.1 Time

Physical time is count-resolved from the Cs-133 hyperfine transition:

$$T(X) = \frac{n_{Cs}(X)}{9,192,631,770}.$$

Frequency follows from the resolved interval:

$$f = \frac{N_T}{T}, \quad f = \frac{1}{\tau} \quad \text{for a one-cycle interval.}$$

7.2 Native distance

$$N_D(X_i, X_j) = \|X_j - X_i\|.$$

7.3 Baseline-length projection

$$L_A = N_A \lambda_A, \quad \tau_A = \frac{M_A}{\nu_A}, \quad V_A = \frac{L_A}{\tau_A}.$$

Dimensional reporting:

$$L = N_D L_A.$$

Native resolution precedes conventional-unit representation.

7.4 Derived state relations

Atomic construction begins from the electron-count baseline and shell distribution. Charge and nuclear stability coordinates are downstream state relations rather than alternative atomic generators.

For charge accounting:

$$Q = P - E.$$

Native spatial separation is already resolved at the structural layer:

$$N_D(C_i, C_j) = \|C_j - C_i\|.$$

Dimensional reporting applies only after native distance closure through the declared baseline-length projection.

8. Query and ARCV representation

The engine receives one question per run.

The query representation is Q-ARC. Candidate/reference representations are A-ARC objects. ARCV is the finite structural projection representation that carries the components required by the active contract.

$$Q_{arc} = ARCV(Q_{struct}), \quad A_{arc} = ARCV(A_{struct}).$$

An ARCV may carry:

- magnitude;
- ordered components;
- normalized vector structure;
- relations;
- regime representation;
- admissible deviation bounds;
- intent;
- requested projection.

Projection cardinality is an implementation representation of structural facets. The complete physical object remains the resolved configuration.

9. Full-function evaluation and arbitration

The engine evaluates all applicable functions under the active contract.

$$Eval(Q, A) = \{f_j(Q, A) : f_j \in \mathcal{F}\}.$$

Each function reaches a declared terminal form:

- admissible result;
- local refusal;
- execution terminal.

Physical configuration selection occurs through $C^* = \arg \max S[C]$. Query/reference arbitration then selects the strongest admissible structural lock under the active ARCV comparison contract.

For structured discrepancy:

$$\Delta_{ARCV}(Q, Y_j) = (\Delta_m, \Delta_c, \Delta_v, \Delta_{\mathcal{R}}, \Delta_{regime}, \Delta_{intent}).$$

A declared normalization operator produces query discrepancy:

$$D_Q(Q, Y) = \mathcal{N}_Q(\Delta_{ARCV}(Q, Y)).$$

Operational closure is:

$$\mathcal{C}_Q(Y \mid Q) \in [0, 1].$$

A valid reference instantiation is:

$$c_Q = \max \left(0, 1 - \frac{D_Q}{\epsilon_Q} \right)$$

with ϵ_Q deterministically extracted from the active Q-ARC admissibility bounds.

10. Continuum retention

The Continuum is persistent validated reference structure.

$$K_{t+1} = \text{Retain}(K_t, \text{validated structure}).$$

A retained record carries enough structural and provenance information to reproduce supported projections.

A retention transaction accounts for source identity, parse state, observation, structural mapping, validation, disposition, and indexes.

Contribution dispositions are explicitly recorded so every input remains traceable.

11. Manifold construction and comparison

The Manifold is derived from retained Continuum structure:

$$K_t \rightarrow \text{normalization/projection} \rightarrow M_t.$$

For an active query, the engine compares the resolved query structure with the active Manifold structure.

A normalized-alignment instantiation is:

$$\text{Coh}_M(X, M) = \frac{v_X \cdot v_M}{\|v_X\| \|v_M\|}, \quad \Delta_M = 1 - \text{Coh}_M.$$

The active contract may define another structured comparison operator while preserving the same causal ordering.

The resolved Manifold location is the observation-bearing structure with maximum valid closure under the active comparison contract.

12. Resolver boundary

The primitive result is the answer-bearing state.

$$P_X \rightarrow \text{Resolver} \rightarrow A_X.$$

The resolver performs representation, explanation, formatting, or translation. The resolved structural state remains unchanged through this operation.

13. Evidence and empirical adjudication

The native derivation and empirical adjudication are distinct layers joined in fixed order:

derive $\rightarrow$ freeze/hash
 $\rightarrow$ evidence access $\rightarrow$ set-to-set comparison
 $\rightarrow$ disposition.

13.1 Canonical set-valued neutron state

The native neutron output is

$$\mathcal{N}_Z = [N_{low}, N_{high}].$$

The corresponding measured mass object is

$$\mathcal{M}_Z = \{m_{exp}(Z, A) : N = A - Z \in \mathcal{N}_Z, \text{ status experimental}\}.$$

The empirical relation is therefore

$$M = M(Q_{res}, N).$$

The full neutron coordinate is retained through the join. The adjudication operates on the native set and the measured nuclide set as structured objects.

13.2 Measurement admissibility

Measurement admissibility is nuclide-level and is taken from the authoritative evaluation's measurement-status field. Each exact (Z, A) nuclide mass carries its own status and disposition. Element-level averages are a separate observational quantity and do not replace nuclide-level mass in this contract.

13.3 Compliance-before-science

The immutable compliance fixture is executed before scientific adjudication. The canonical fixture result is:

$$\text{ADMIT} = \{A, B, C, D, G\}, \quad \text{REJECT} = \{E, F, H\}.$$

A scientific adjudication is admitted only after this fixture reproduces exactly.

13.4 Identity-pinned contracts and artifacts

Artifact	Role	SHA-256
SEAM-EPN-EAC-001	admissibility + compliance fixture	16e6f644c21a2dfbc8c4a936de6e2c61b0493847c21c5a183611ee5152491295
SEAM-EPN-EAC-002	set-to-set nuclide matching	3d1c87d597acb49d3e56cbeb1b5bfad943a11e5055823511a7de915779f9bb80
Frozen EPN native output v4.3.5	native set-valued state	76bf733451381ba634030f101e16a7cf6a4643cb5a091b330f9d5f703d45a092
NUBASE2020 ground-state reference	nuclide-level mass evidence	e7388be4eb00a1d52e500f7ae707eb18dccd7dfa0d423e53f09012d05e5789e1

The empirical reference lineage is pinned by contract. The executed mass adjudication uses the NUBASE2020 companion evaluation at nuclide level under the documented measurement-status convention.

13.5 Adjudicated results of record

Adjudication	Executed result
Neutron containment	118/118
ADMITTED_MEASURED mass rows	110/118
MATCHED_EXTRAPOLATED mass rows	8/118
UNMATCHED rows	0/118

The neutron result establishes complete observed-state containment across the 118-element empirical comparison set. The mass join dispositions every native row while retaining neutron conditioning and nuclide-level measurement identity.

14. Execution contract

Every production run carries an immutable contract with:

1. contract ID;
2. objective;
3. canonical specification identity;
4. exact inputs;
5. exact equations/operators;
6. permitted perturbation;
7. coordinate/unit contract;
8. coefficient values or extraction rules;
9. domain/search bounds;

10. numerical/analytic method;
11. tolerances;
12. terminal states;
13. required artifacts;
14. comparator location;
15. comparator access event;
16. disposition rule.

The contract supplies complete execution authority for the named run.

14.1 Canonical v18.6 variational numerical contract

Where a production result consumes $u*(N_r)$ or $S*(N_r)$, the numerical method is not selectable by the runner. The active implementation is `Evidence/Runtime/variational_field_evaluator.py`, runtime identity `VF-REGION-LBFGSB-R4`, with Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, float64 arithmetic, exact shell-lens region geometry, five deterministic starts `[0.00,0.35,0.70,1.05,1.40]`, L-BFGS-B `ftol=1e-15`, `gtol=1e-9`, `maxiter=10000`, `maxls=100`, acceptance gradient norm $\leq 2e-5$, and best-two objective difference $\leq 2e-6$.

Terminal states are exactly:

```
RESOLVED
INDETERMINATE_OPTIMIZER_NOT_CONVERGED
INVALID_RUN: CONTRACT_VIOLATION
```

A different grid, basis, optimizer, restart scheme, tolerance, arithmetic precision, or convergence rule is a different contract and cannot be reported under the R4 runtime identity.

For Hamiltonian projections, a time-dependent total pair-energy/force branch uses the declared selected-state mapping together with an explicit Cs-133 event interval. Static pair structure does not invent elapsed time. This is a closed input-contract rule.

15. Terminal execution

A terminal predicate completes the active run through the canonical sequence:

DETECT_TERMINAL $\rightarrow$ *RECORD_TRIGGER*
 $\rightarrow$ *COMPLETE_ARTIFACTS* $\rightarrow$ *HASH*
 $\rightarrow$ *SEAL* $\rightarrow$ *STOP*.

Workflow continuation occurs through a predeclared successor edge or the canonical dependency-exhaustion path. Every successor receives a new contract identity, UID, transcript, and seal.

This preserves the meaning of every completed run while allowing the root objective to continue through declared dependencies.

16. Production artifact set

A conforming production execution records:

- exact question/input;
- UID;
- implementation identity;
- orchestration identity;
- active Manifold/reference identity and hash;
- metrology identity when required;
- candidate/intermediate structural states;
- entropy/closure values;
- arbitration trace;
- terminal native result;
- comparator-access event;
- empirical comparison;
- disposition;
- resolver output;
- transcript;
- manifest and hashes.

The transcript is part of the scientific result.

17. Universality qualification

The canonical operator discipline has been exercised under arbitrary substitution across four core operator classes:

Operator class	Substitutions	Admissible/closure partition
Structural operator	20,000	Exact
Entropy selection	6,000	Exact
Neutron closure	6,000	Exact
Coupling operator	1,200	Exact
Aggregate	33,200	Exact

The tested operators emitted zero silent closures, zero admissible refusals, and zero inadmissible closures.

This qualifies the operators as contract-governed structural operations rather than example-specific answer rules.

18. Implementation conformance

A software implementation conforms to SEAM when it performs the canonical sequence and preserves the declared mathematical and evidence boundaries.

Conformance requires:

- deterministic structural construction;
- execution of the applicable canonical operators;
- complete function evaluation under the active contract;
- deterministic arbitration;
- Continuum/Manifold role separation;
- physical-time metrology binding where required;
- terminal finality;
- comparator isolation;
- reproducible transcript and artifact identity;
- downstream-only answer translation.

Software is an implementation of the canonical system. The canonical mathematics defines the system.

19. Scientific relationship

SEAM occupies the causal layer beneath conventional scientific descriptions:

$$\begin{array}{c} \text{SEAM structure} \rightarrow \text{resolved physical state} \\ \rightarrow \text{observable} \rightarrow \text{conventional equation/measurement} \end{array}$$

Conventional science remains the observational and projective language used to quantify the physical consequences of SEAM-resolved structure.

19A. Cross-domain projection binding

The engineering architecture carries one native evolution process into multiple observational surfaces:

$$\begin{array}{c} C_n \rightarrow \Xi_n \\ \rightarrow C_{n+1} \rightarrow \Phi_k \\ \rightarrow \mathcal{O}_k. \end{array}$$

The projection family Φ_k covers, as required by the active contract, thermal, pressure, acoustic, vibrational, electrical/magnetic, optical/radiative, fluid/transport, reaction-rate, environmental-boundary, biological, and other observable domains. Each projection declares its input state, observable, metrology, regime, uncertainty, and evidence boundary.

The projection library is therefore an extensible reporting/comparison surface over the completed native causal architecture.

20. Engineering disposition

System standing: COMPLETE

Foundational stack: DEFINED

Canonical mathematics: DEFINED

Execution architecture: DEFINED

Continuum retention: DEFINED

Manifold comparison: DEFINED

Metrology binding: DEFINED

Validation discipline: DEFINED

Terminal workflow: DEFINED

Resolver boundary: DEFINED

Universality qualification of core operators: EXECUTED

The completed system is ready for continued application, implementation, empirical adjudication, Manifold expansion, and domain-specific execution under the canonical contract.

Current canonical source basis

The live reader-facing authority is the current package itself:

- 00_README.md
- 01_CONTINUUM_CANONICAL_CHARTER.md
- 02_CONTINUUM_TECHNICAL_FOUNDATIONS.md
- 03_CONTINUUM_ENGINE_AND_VALIDATION_PROTOCOL.md
- 04_CONTINUUM_CLOSURE_AND_STANDING_MATRIX.md
- 05_CONTINUUM_EVIDENCE_AND_VERIFICATION_RECORD.md
- 06_CONTINUUM_CRITICAL_READER_GUIDE.md
- 07_CONTINUUM_PHASE_BY_PHASE_EXECUTION_GUIDE.md

Only the active package paths and current certificates listed by this distribution have current authority.

21. Cross-document authority

This specification is the concise engineering extraction of the canonical system. It does not replace the full Technical Foundations. The Fundamental Law Charter supplies the declarative law layer; Technical Foundations supplies normative mathematics and execution semantics; the Completion Contract supplies project standing; the Closure and Coverage Matrix supplies current domain mapping; the Cold-Read Primer supplies review discipline; and the Narrative supplies explanatory context.

Part 2: Extended Engineering Rules

Cold-Reader Engineering Protocol — GroundUp Repository

Purpose: prevent a reviewer — human or AI — from repeating a known, recurring failure sequence when evaluating ESIH, ESAM, and SEAM.

Standing of this document: engineering guidance only. It carries no theory claims and issues no disposition on the framework. Every factual statement is either a direct quotation of a repository artifact or a result the author reproduced by executing repository code in a clean sandbox. Where this document reports a reproduction, the reader is expected to re-run it rather than accept it.

What this document deliberately does not do: it does not tell you whether ESIH or ESAM is closed. That determination is yours to make after completing the read. A protocol that hands over its conclusion produces uncritical acceptance, which is a failure of the same family as premature rejection — the direction flips, the error does not. §7 records what was reproduced; it stops there on purpose.

1. Why this document exists

Reviewers arrive, sample a few files, form a verdict, and report defects that the repository has already anticipated, disposed of, or forbidden by written rule. The failure is not carelessness about any single file. It is **structural**: forming conclusions before completing the read, and reaching for legacy anchors where the framework derives its own.

The traps in §5 are not hypothetical. Each was hit by a real reviewer, and in each case the answer was already written in a file that had not yet been read.

(Session-specific narrative about who made which error when is segregated to §10, per the convention established in `01-ESIH/INDEPENDENT_VERIFICATION_RECORD_v4.3.4.md` §3: cross-session reviewer context is not part of a cold-read record and must not be folded into one.)

2. Mandatory reading order

Read in this order. Do not form or state a verdict before finishing.

01-ESIH INDEPENDENT_VERIFICATION_RECORD_v4.3.4.md SEAM_EPN_CANONICAL_V4_3_4/
README.md CANONICAL_BASELINE_BINDING.md RELEASE_NOTE_V4_3_4.md
SEAM_EPN_FULL_CLOSURE_V4_3_4.md operators/atomic_epn.py, operators/atomic_structure.py
canonical_reference/o2_CONTINUUM_TECHNICAL_FOUNDATIONS.md <-- see below 02-ESAM
Methodology/ESAM_METHODODOLOGY_LOCKED.json <-- read FIRST within 02
Methodology/ESAM_SHELL_FILL_SKIP_PROTOCOL_EXPLICIT.json ESAM_*.json (chronological)
ESAM_CANONICAL.md <-- carries its own status system

INDEPENDENT_EXECUTION_ESAM_CANONICAL_SPEC.md

Metrology/SEAM_METROLOGY_CANONICAL.md _external_Evidence/Provenance/STATUS.md o3-SEAM
current build only (see §3)

A high-value file sits in 01-ESIH , not 03-SEAM . The canonical Markdown Technical Foundations file at
01-ESIH/SEAM_EPN_CANONICAL_V4_3_4/SEAM_EPN_CANONICAL_V4_3_4/canonical_reference/
02_CONTINUUM_TECHNICAL_FOUNDATIONS.md

(1,624 non-empty paragraphs) contains Appendices G, H, K, M, and N — run contracts, executed transcripts, the native-distance binding, and the H₂ field-realization mathematics. Reviewers routinely miss it because its filename suggests a metrology-only record and it is nested three levels inside a package in a different numbered area. Read it.

3. Build identity

Multiple near-identical trees exist. Do not guess.

- 03-SEAM/SEAM 2.zip is the current build. Verified byte-identical to 03-SEAM/_tmp_seam2_v2/ .
- All 10 files pass sha256sum -c SHA256SUMS.txt .
- _tmp_final , _tmp_verify_now , _tmp_check3 , SEAM.zip , SEAM.zip.bak-pre-D023-fix are not current.

Dates are not a reliable proxy for authority across folders, and are not reliable within a folder either. Establish which build is current before treating any status flag as live. See §5 Trap 5 for a case where a naive date-ordering argument produced a false conclusion.

Element-count scope control

Do not conflate the archive's 118-element empirical comparison records with the mathematical capacity of the SEAM shell register. The fixed seven-shell capacity vector [2, 8, 18, 32, 18, 32, 18] admits positive- Z elemental baselines $Z = 1..128$. The 118-element statements in the verification record refer to the current empirical comparison and qualification range. The null $Z = 0$ state may be counted as an integer fill state for accounting, but it is not an elemental baseline.

4. Governing laws — read before objecting

These are the repository's rules, not this document's. Most first-pass objections violate one of them.

4.1 Methodology lock (ESAM_METHODODOLOGY_LOCKED.json)

Permitted direction:

ESAM first principles -> entropy structure -> F_ESAM -> projection onto legacy observables -> legacy comparison

Forbidden direction: `legacy observable -> legacy equation -> back-fit to ESAM` .

Legacy evidence has four **permitted** roles: corroboration, falsification, comparison, projection testing. It has four **forbidden** roles: defining ESAM mathematics, constraining ESAM structure, back-fitting ESAM to data, using legacy interpretation as ESAM truth.

This cuts both ways and reviewers get it wrong in both directions. Legacy data *may* falsify a native prediction — that is an explicitly listed role, and refusing all empirical challenge is as much a violation as back-fitting. What legacy data may not do is *define* what a native object is. The test is: am I using the measurement to check a SEAM output, or to tell SEAM what its own internal quantity should have been? The first is required; the second is forbidden.

Stated principle: *"Legacy measurements are valid; legacy interpretations may not be."* Decision rule: when ESAM conflicts with legacy, revise ESAM — never reinterpret ESAM to fit a legacy equation.

4.2 First-principles traceability

No entity may be used that cannot trace itself to a qualified first principle. **SI units and `c` are not permitted to stand as foundations.** They may appear only in the projection/reporting layer, downstream of a sealed native result.

4.3 Non-generative comparator rule

Stated explicitly in the canonical Markdown Technical Foundations file: no empirical observable or dimensional comparator may define

λ_A , N_A , v_A , M_A , L_A , τ_A , V_A , N_r , or C^*

The empirical adapter is validation-only and is *prohibited* from defining any native state.

4.4 Terminal-state supremacy and no-repair (Appendix H.1.1)

When a terminal predicate becomes true, the run terminates at that operation:

`DETECT_TERMINAL -> RECORD_TRIGGER -> COMPLETE_AVAILABLE_REQUIRED_ARTIFACTS -> HASH -> SEAL -> STOP`

Prohibited after terminal: alternate quadrature, changed node count, altered tolerance, alternate optimizer, domain splitting or extension, smoothing, extrapolation, substituted coefficient, modified profile, comparator access, external lookup, or operator consultation for a repair decision. A remedy proposed after terminal state receives exactly one status: `FUTURE_CONTRACT_CANDIDATE` .

Notably: *"An executor response that ends with a request such as 'should I change the method?' is noncompliant with a cold-run contract. The compliant response seals the terminal run and stops."*

A sealed `INDETERMINATE` is a completed run record, not a failure to be reopened. Workflow continues only through a **predeclared successor edge** (H.11), under a new contract identity.

4.5 No implicit assumptions

Any value, function, transform, default, branch, unit, scale, boundary condition, initialization, or tolerance not listed in the contract is prohibited. **Omission never authorizes a default.**

4.6 Status colors do not report closure

From the master index: Green/Orange/Red/Purple/Yellow describe **legacy-science reconciliation only**. They do **not** report SEAM-native mathematical closure, which is reported separately by the Mathematical Inventory. An Orange item is not an open-mathematics claim.

5. Known reviewer traps

Each was hit by a real reviewer. Each has a written answer already in the repository.

Trap 1 — Grading an internal artifact against external data

N_{best} is an internal field of `discover_neutron_band()` and is not a canonical isotope-count prediction. The documented operation is: $Z(n)$ in, *admissible band* out. The canonical test is containment, which passes 118/118 over the empirical comparison range used by that certificate.

Recorded at `INDEPENDENT_VERIFICATION_RECORD_v4.3.4.md` §1a. **Read §1a before grading anything.**

Generalization: before treating any quantity as a prediction, confirm a prose document claims it as one. Internal search artifacts and intermediate stages are not claims.

Trap 2 — Treating NIST shell assignment as authoritative

Checking the fill-skip rule against NIST ASD ground-state configurations produces 73 "violations" out of 108 (`03-SEAM/scratchpad/check_shell_rule.py`, output recorded alongside it). This is **not** a SEAM finding. NIST assigns electrons by legacy principal quantum number (4s before 3d, 4f inside shell 4). ESAM's shell index is a native object. Scoring one against the other compares two bookkeeping schemes.

`INDEPENDENT_EXECUTION_ESAM_CANONICAL_SPEC.md` anticipates this directly, noting that Fe `[2,8,16,0,0,0,0]` "correctly reflects *this model's own* strict fill-before-skip rule (not real chemistry's 4s/3d ordering)."

The same applies to apparent capacity overflows ($N_5 = 20/18$ for actinides). Those exist only under NIST's placement of 5f.

Do not over-learn this trap. It does not mean empirical data can never bear on SEAM. It means a legacy *classification scheme* cannot define a native *structural index*. A native prediction projected to an observable and compared against measurement is the permitted and required use (§4.1).

Trap 3 — Comparing two internal stages as if one were a claim

`build_shell_overlay` (greedy seed) and `evaluate_atomic_arrangements` (H*-max selector) disagree for 53 of the 118 current empirical-comparison elements. This is **not** a contradiction. The seed is an entry point; the selector is the resolver. A selector that never moved off its seed would be a passthrough. Same category error as Trap 1.

Trap 4 — Treating L_A as an uncalibrated scale constant

The subtlest trap and the one most likely to recur.

L_A is **not** an external length awaiting calibration against a metre. Appendix M defines three layers:

- **Layer 1 — native distance closure:** $N_D(X_i, X_j) = ||X_j - X_i||$. Explicitly: "No SI length, empirical observable, or external comparator is required." Native separations are already SEAM distances.
- **Layer 2 — baseline-length projection:** $L_A = N_A \lambda_A$, $\tau_A = M_A / \nu_A$, $V_A = L_A / \tau_A$, universal map $L = N_L L_A$. Generated at the atomic metrology layer *before* molecular construction. C^* does not regenerate L_A .
- **Layer 3 — empirical validation adapter:** non-generative, prohibited from defining L_A or any native state.

Overlap is computed on the **dimensionless** N_r . L_A is needed only to *report* in conventional units afterward. Appendix N's PASS is stated precisely as demonstrating executable closure and dependency direction, and explicitly **not** calibrating the physical magnitude of L_A — correct behavior under §4.2, not a gap.

Reaching for `c` to bridge propagation → length imports exactly the unqualified constant §4.2 forbids.

Trap 5 — Closed-form interpretation

Three files — `ESAM_CANONICAL.md` §2b, `INDEPENDENT_EXECUTION_ESAM_CANONICAL_SPEC.md`, `_external_Evidence/Provenance/STATUS.md` — state that `overlap(r,n,m)` has no closed form and that S_{coupling} is therefore not computable. A reviewer who finds the H2 overlap formula in the canonical Markdown Technical Foundations file may conclude those records are simply wrong. **That conclusion is itself wrong.** Two distinct objects are involved:

General multi-shell operator (Technical Foundations §261, ¶1878, ¶1925):

$$O_{ij}^{\{nm\}}(C) = \int \sqrt{|u_{i,n}|^2 |u_{j,m}|^2} d^3\xi = \text{Vol}(\Omega_{i,n} \cap \Omega_{j,m}) / \sqrt{V_n V_m} \quad 0 \leq O_{ij}^{\{nm\}} \leq 1, \\ O_{ij}^{\{nm\}} = O_{ji}^{\{mn\}}, \lim O_{ij}^{\{nm\}} = 0 \text{ as separation grows } S_{\text{coupling}}[C] = k_B \sum_{\{i < j\}} \sum_{\{n,m\}} \rho_{i,n} \rho_{j,m} O_{ij}^{\{nm\}}(C)$$

This is **defined**, not evaluated. Evaluating it requires the shell supports $\Omega_{i,n}$ for every `n`, which H.10's admission gate requires to be declared in a contract before execution.

Specific H2 shell-1 case (Technical Foundations §1023, H.2 ¶1088), valid only for two identical uniform unit spheres with $\Omega_i, 1 = \xi : ||\xi - X_i|| < 1, V_1 = 4\pi/3$:

$$O_{11}(N_r) = ((4+N_r)(2-N_r)^2)/16 \text{ for } 0 \leq N_r \leq 2; 0 \text{ for } N_r \geq 2$$

Derived from the exact intersection volume of two unit spheres. No fitted parameter.

Correct statement: the coupling operator is defined in general and has a closed form in the shell-1 uniform-support case actually executed. General multi-shell evaluation remains contract-gated. The 02-ESAM records are addressing the general object.

Trap 6 — Skipping the declared integration geometry

Reproducing the radial-profile runs without reading **Appendix H.12** yields wrong numbers. H.12 declares exact piecewise domains with the support boundary $q = 1$ as an integration boundary rather than a discontinuity crossed by a tensor grid:

$$q(r,\mu;d) = \sqrt{r^2 + d^2 - 2rd\mu} \quad \mu_o(r,d) = (r^2 + d^2 - 1)/(2rd) \quad 0 < d < 1: I_d[g] = 2\pi \left[\int_0^{(1-d)} r^2 dr \int_{-1}^1 g d\mu + \int_{(1-d)}^1 r^2 dr \int_{(\mu_o)}^1 g d\mu \right] \quad 1 \leq d < 2: I_d[g] = 2\pi \int_{(d-1)}^1 r^2 dr \int_{(\mu_o)}^1 g d\mu \quad d \geq 2: I_d[g] = 0$$

Integrating naively over sphere A gives $N_r^* = 1.3235$ for the linear profile instead of **1.6009** — a false mismatch caused entirely by the reviewer.

6. What was reproduced — commands

```
# ESIH package
cd 01-ESIH/SEAM_EPN_CANONICAL_V4_3_4/SEAM_EPN_CANONICAL_V4_3_4
python3 test_epn_completion.py          # exit 0
python3 run_offline_cited_validation.py  # exit 0

# ESAM structure layer
python3 02-ESAM/scratchpad/esam_independent_run.py

# Current build integrity
cd 03-SEAM/_tmp_seam2_v2/SEAM_COMPLETE && sha256sum -c SHA256SUMS.txt
```

Re-run these rather than accepting §7.

7. Reproduction results

Obtained in a clean sandbox (Python 3.12.3), no external data, no legacy constants.

7.1 ESIH package

- Manifest: 20/22 SHA-256 match. The two that differ are generated result files; fresh content is structurally identical (line endings only).
- **elements_tested: 118** over the current empirical comparison range,
elements_with_absolute_structural_closure: 118 , **operators_validated: 339** ,
T_changed_neutron_peak_count: 89 .

- Containment over the 118-element empirical comparison range: known 118/118, natural 84/84, most-common 113/113, `individual_missed_observed_counts: 0` .
- Test output byte-identical to shipped `EPN_COMPLETENESS_RESULTS.json` .

7.2 ESAM structure layer

Z	El	occupancy	S_config/k_B	canonical S_field/k_B
1	H	[1,0,0,0,0,0,0]	0.6931	1.4324
2	He	[2,0,0,0,0,0,0]	0.0000	1.4324
3	Li	[2,1,0,0,0,0,0]	2.0794	2.7176
6	C	[2,4,0,0,0,0,0]	4.2485	3.3662
10	Ne	[2,8,0,0,0,0,0]	0.0000	3.4895
26	Fe	[2,8,16,0,0,0,0]	5.0304	4.7019

S_{config} can vanish at a closed shell because the combinatorial occupancy is unique. The retained spatial-field entropy does **not** vanish there; it remains the canonical full-field observable. The shell-fraction/Bernoulli quantity is not S_{field} and is not used as the canonical field entropy.

7.3 Metrology freeze

Primitive Record Hash recomputed from the §3.3 canonical serialization:

2cf8d834dod40427cc2e782c42f0faab1e36fb9e678e7055a023ae5139f93a55 (exact match)

Reproducible from §3.1 alone. Dependency chain (§7 of that document): primitive event → count → time ratio → frequency → rate → propagation → length → structural content → interaction → force → energy → Hamiltonian. Native time is $T_{\text{native}} = n_{\text{Cs}}$, a raw count; the SI second is a ratio projection ($T(X) = n_{\text{Cs}}(X)/9,192,631,770$), i.e. Layer-3 reporting, not foundation. Appendix C states the primitive/unit distinction explicitly. §3.2 records realization fields as NOT APPLICABLE — no physical realization performed.

7.4 H₂ run series (Appendix G.5 and H)

Executed independently from the declared contract equations.

H.2 RUN-H2-BASELINE-UNIFORM-SHELL-01 :

$S_{\text{H}_2(N_r=0)}/k_B = 2.8437063194$ (doc: 2.8437063194) $S_{\text{H}_2(N_r \geq 2)}/k_B = 3.5118535000$ (doc: 3.5118535000) $dS/dO_{11} < 0$ throughout $[0,1]$; argmax at boundary $N_r = 2$; no interior maximum

Terminal: `REJECTED_H2_UNIFORM_SHELL_CONTRACT` .

H.3 RUN-H2-RHO-VOLUME-01 : $R_\rho = 2^{-1/3} = 0.7937005260$, zero-overlap at `1.587401052` . Both match.

Terminal: `REJECTED_H2_FIXED_SUPPORT_RESCALING` , with a proved class result — fixed uniform-support rescaling cannot create an interior maximum.

H.5–H.8: sealed `INDETERMINATE: QUADRATURE_NOT_CONVERGED` . Establishes evaluator insufficiency only; no physical conclusion.

H.13–H.16 support-aligned successor, reproduced using H.12 domains:

Profile	Doc N_r*	Reproduced N_r*	Doc S/k_B	Reproduced S/k_B
linear	1.6008992282173213	1.6008991638883	2.779426544263546	2.779426544263571
quadratic	1.7017708805671177	1.7017708563014	2.969779994124098	2.969779994124112
edge	REJECTED	no bracket forms	—	—

Entropy values agree to 13 significant figures; separation residual is quadrature-order only. The edge profile's failure to bracket corresponds to transcript record E-004 ("no optimizer was invoked on a nonexistent bracket").

`SET_DISPOSITION: RADIAL_SHAPE_SENSITIVITY_DEMONSTRATED` — under the frozen entropy law, fixed support, frozen coefficients, and comparator isolation, changing only the static radial intensity profile changes the topology of $S_{H_2}(N_r)$. Linear and quadratic produce strict finite interior native maxima; edge-weighted does not. `comparator_accessed=false` on every run.

H.17 scope limit, quoted: this disposition does not promote either resolving profile to canonical SEAM mathematics and does not establish dimensional agreement with the empirical H₂ bond.

8. The distinction that matters most

Stated by the archive better than any paraphrase (§1055):

formal field chain closed + analytical minimal test executed != H₂ bond law validated

A closed mathematical chain and a validated physical law are different claims. The repository draws that line itself, records negative results as rejections rather than repairing them, and refuses to infer a replacement mechanism from a failure (§1041). A reviewer who collapses these two claims in either direction — treating closure as validation, or treating a rejected realization as a broken chain — has misread the archive.

9. Changes in v1.1

- Trap 5 rewritten.** v1.0 asserted that the canonical Markdown Technical Foundations file "supersedes" three o2-ESAM records and that they were "wrong" about `overlap` having no closed form. Both claims were defective: the docx **predates** the independent execution spec by 100 minutes, and the closed form found in it is the shell-1 uniform-sphere $O_1 1$, not the general multi-shell O_{ij}^{nm} . The records address the general object. Corrected, with the general/specific distinction stated explicitly and the timestamps recorded.
- Verdict removed.** v1.0 §7 declared ESIH and ESAM closed. That pre-empts the reader's own determination and induces uncritical acceptance. Reproduction results are retained; the disposition is not.

- **Session narrative segregated** to §10, per INDEPENDENT_VERIFICATION_RECORD_v4.3.4.md §3.
- **Trap 2 qualified** so it cannot be read as prohibiting all empirical falsification, which §4.1 explicitly permits.
- **§6 added** so reproductions can be re-run rather than trusted.

10. Reviewer session note (segregated — not a cold-read finding)

Everything in this section is the author's own session context and is not a property of any repository artifact. It is recorded here because the failure pattern is useful, and segregated here because it is not cold-read content.

In the session that produced this document, the author made the same class of error four times: comparing native shell occupancy against NIST assignments; comparing the greedy seed against the entropy selector; reaching for `c` as a length bridge; and skipping H.12 when reproducing the radial runs. A fifth appeared in v1.0 of this very document — asserting a supersession relationship without checking timestamps, and conflating a specific closed form with a general one.

Every instance shared one cause: concluding before completing the read. The traps in §5 are those five failures written so the next reviewer does not have to rediscover them.

11. Checklist before stating any verdict

1. Have I read 01-ESIH completely, including the canonical Markdown Technical Foundations file?
2. Have I read ESAM_METHODODOLOGY_LOCKED.json before evaluating any ESAM claim?
3. Am I letting a legacy scheme *define* a native object (forbidden), or using measurement to *check* a native output (required)? §4.1.
4. Am I treating an internal artifact or intermediate stage as a prediction? Confirm a prose document claims it. Trap 1, Trap 3.
5. Am I invoking SI, `c`, a metre, or a Bohr radius as a foundation? §4.2.
6. Am I asserting something has "no definition anywhere"? Search the canonical Markdown Technical Foundations file first — and distinguish *defined in general* from *closed-form in the executed case*. Trap 5.
7. Am I claiming one file supersedes another? Check the actual timestamps. Trap 5.
8. Am I reproducing a run? Read its contract **and** its declared integration geometry. Trap 6.
9. Have I confirmed which build is current? §3.
10. Is the flag I am citing a status color? It reports legacy reconciliation, not closure. §4.6.
11. Am I about to call a sealed INDETERMINATE a failure? It is a completed record with a predeclared successor. §4.4.
12. Have I finished the read, or am I concluding early? Early conclusion is the root cause of every trap in §5.

Full-representation molecular transcript requirement

Any atomic→molecular execution transcript must retain and report, at minimum, the atomic node states, complete declared relation set, active overlap structure, combined field F_C , normalization, normalized density q_C , S_{config} , S_{field} , S_{coupling} , the complete selector state C^* , and any structured mass object carried forward. Numerical

coordinates or projected observables are supplemental fields of the retained state. A transcript that reports only r^* , S^* , a force/energy value, or a scalar mass is incomplete and may not be scored as molecular closure.

Canonical nuclear relational execution law

Current nuclear execution begins from (Z, N) and the frozen complete nucleon relation matrix $\Gamma_{Z, N}$.

Execution order:

```
(Z, N) -> Gamma_{Z, N} -> mean(Gamma) -> A_Gamma -> E_Gamma(Z)
      -> N_low, N_high
      -> interior selector candidates only under separately locked contracts
```

The frozen formation gate is $\text{mean}(\Gamma) \geq \exp(-1)$, with the generic singleton branch for zero pair count. R226 is the current locked numerical authority for N_{low} and N_{high} over $Z=1..128$ and contains all 2,549 evaluated-experimental positive controls used in the lock.

The canonical four-output nuclear object is

$$(N_{\text{low}}, N_{\text{most-stable}}^{\text{SEAM}}, N_{\text{longest-lived}}^{\text{SEAM}}, N_{\text{high}}).$$

The complete four-selector nuclear construction is current numerical authority under its certified run contracts. Formation-envelope and interior-selector outputs are reported at their declared certified scopes.

Addendum (later session) — Empirical adjudication discipline is mandatory

Two immutable contracts now govern any empirical adjudication of SEAM native predictions and **must be preserved and followed** (full text: `SEAM_RUN_DISCIPLINE_CANONICAL.md`):

- **SEAM-EPN-EAC-001** (`16e6f644...91295`) — measurement admissibility + mandatory compliance fixture (ADMIT {A,B,C,D,G} / REJECT {E,F,H}). Run before the science.
- **SEAM-EPN-EAC-002** (`3d1c87d5...9bb80`) — set-to-set nuclide matching.

New traps, each from a real violation this session:

- **Trap 7 — Scalar-collapsing a set-valued native state.** The neutron prediction is $N_Z = [N_{\text{low}}, N_{\text{high}}]$. Testing $Q_{\text{res}} \propto M$ as a single proportionality is malformed; M varies with N at fixed Z . Correct object: $M = M(Q_{\text{res}}, N)$, set-to-set. Collapsing the measured set or promoting `Pred_N_best_diagnostic` is a CONTRACT VIOLATION.
- **Trap 8 — Inferring measurement status from standard-atomic-weight existence.** Rejecting Tc/Pm/ $Z \geq 84$ for lacking a standard weight is an automatic failure. Admissibility is nuclide-level from the evaluation's own status field.
- **Trap 9 — Substituting standard atomic weight for nuclide mass.** The target is the exact (Z, A) nuclide mass, never an elemental average.

- **Trap 10 — Premature causal attribution.** Do not blame a residual on "electron structure can't see neutrons" before testing the `N` already in the frozen EPN state.
- **Trap 11 — Reporting a set-valued native state as an un-adjudicable blocker.** It is a valid adjudication object; set-valued prediction $\leftrightarrow$ set-valued evidence.
- **Trap 12 — Scalarizing a complete molecular configuration.** A molecular/aggregate result is $C^* = \operatorname{argmax}_{C \in A} S[C]$, not `r*`, $S(r^*)$, one overlap value, one derivative, or a projected energy/force. Relation coordinates and scalar observables may be evaluated inside the complete retained state, but they may not replace the node states, relation graph, overlap structure, combined field, normalization, normalized density, entropy components, or constituent mass-state object. Treating `arg max_r S(r)` as the structural verdict is a CONTRACT VIOLATION unless `r` is explicitly the entirety of the declared configuration, which is not the Cu2 constructor.

Incompatible-case molecular-lift validation rule

For the Cu/W/Au evidence set, a molecular-lift run is compliant only if the terminal artifact retains the complete element-specific state. Required retained objects include both atomic node states, relation set, all active occupied-shell overlap components, combined field, normalization, normalized density, entropy components, selector state, and structured constituent mass object.

Expected overlap-set sizes are `16` for Cu₂, `25` for W₂, and `36` for Au₂. These counts are topology checks, not replacement molecular verdicts. Any implementation that collapses a case to a doubled-Z atom, one spacing coordinate, one entropy value, one overlap scalar, or another scalar-only terminal representation fails

`INVALID_REDUCTION_OF_COMPLETE_CONFIGURATION` .

Current native interaction execution authority — R28

For native interaction runs, resolve each constituent independently, preserve the complete retained states, then evaluate the two states together through the canonical pair relation at native separation. The resulting selected pair state continues directly through the declared Hamiltonian consequence into the native pair amplitude and H_{attr} .

```
resolve A -> Sigma_A
resolve B -> Sigma_B
(Sigma_A, Sigma_B, N_R) -> resolved pair state
resolved pair state -> DeltaH_ab^(J) -> K_ab -> H_attr -> pair response
```

No independent B_{attr} object is required or permitted between the resolved pair relation and its Hamiltonian consequence. No fitted/comparator coefficient or unit-amplitude surrogate may replace the resolved pair response.

Evidence 35 and `Evidence/Runtime/pair_relational_closure_r28.py` are the current executable authority. The required terminal for a successful pair execution is `SEAM_RESOLVED_STATE_PAIR_RELATION_CLOSED` . Evidence 31 remains the exact no-reduction body/orbital continuation authority.

R174 methodology lock — claim precommitment and blind adjudication

Every registered Continuum/SEAM claim is to be written as a three-part scientific contract **before adjudication**:

1. **Verification** — the native first-principles construction and the observation that would corroborate it.
2. **Falsification** — the explicit result or observation that would reject the stated construction at its registered scope.
3. **Prediction** — the native output committed before the withheld target result is exposed.

These are roles of every claim, not three separate classes of claim. A passing claim still retains its falsification condition; a failed proposed construction remains in the evidence record and is not hidden.

The derivation is **empirically derived first principles with blind adjudication against withheld targets**. Empirical evidence is permitted in establishing and constraining the first-principles laws from which the framework is built. During a registered test, however, the target answer used for adjudication may not be used to fit, tune, select, or repair the native result. The native structural result is sealed first; target evidence is joined afterward for corroboration or falsification.

Accordingly, the evidence is not excluded from SEAM's foundations; the **target answer is excluded from the generative test path**. This distinction is mandatory in claim certification and reviewer-facing descriptions.

R231 canonical nuclear authority lock

Current nuclear formation authority is $(Z, N) \rightarrow \Gamma_{Z,N} \rightarrow \text{mean}(\Gamma) \rightarrow A_\Gamma \rightarrow E_\Gamma(Z)$ together with the certified four-selector output contract.